

The Override Signal

Day 2 · Human and the Loop / Bainbridge's Debt · 25 min · pods of 3-5

THE KNIGHT CAPITAL REENACTMENT (5 MIN)

August 1, 2012. A botched deployment reactivated dormant trading code at Knight Capital. In about **45 minutes** it fired millions of erroneous orders and lost roughly **\$440 million** — nearly destroying the firm. Here's the part that matters for us: during the pre-market deployment the system generated **97 automated alert emails** referencing the fault. They were not configured as critical, no one owned them, and no one acted. A human was, technically, “in the loop.” The loop never ran.

Objective.

Design **one** runtime oversight signal for your pod's case — one that would actually have caught a Knight-style failure. Name what it measures, its threshold, its cadence, who watches it and with what authority, and what it triggers.

A SIGNAL IS OVERSIGHT ONLY IF IT IS...

- Observable** — actually measured and surfaced, not buried in 97 emails.
- Actionable** — a threshold that triggers a specific action, not just a dashboard to admire.
- Owned** — a named role watches it and has the **authority to stop the system**.
- Timely** — it fires fast enough, and before the irreversible step, that judgment can still matter.
- Fatigue- and bias-resistant** — few and prioritized (not alert fatigue), and it watches the human too (not just the model).

THREE SIGNAL ARCHETYPES

- **Override rate** — how often the human changes the agent's output. Too high = the agent is wrong; **too low = rubber-stamping** (automation bias).
- **Drift cadence** — a scheduled look at the *aggregate* of outputs over time, to catch a slow trend no single output reveals.
- **Deskilling proxy** — a periodic check that the human can still do the task *unaided*, so the oversight role stays real.

Your task (≈13 min)

Pick one archetype (or invent one). Specify the signal completely, then test it: **would it have caught Knight?** If a human would still have been buried, redesign it.

FIELD	YOUR ANSWER
Signal — what exactly it measures	
Threshold / trigger	
Cadence — how often / real-time	

Owner + authority — who watches, who can stop it	
Action it triggers	
Would it have caught Knight? Why	

Closing reflection. Knight had a human in the loop and still lost \$440M. What did your signal add that 97 emails didn't? If the answer isn't “surfaced, owned, and actionable in time,” it's another email.

Illustrative for the exercise only. One signal is a starting point, not a monitoring program. Nothing here is legal or regulatory advice.